

Active Directory & IT Help Desk Home Lab Portfolio

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IT Support • Active Directory • Windows Server • Jira Service Management

Built using Windows Server 2022, Windows 10 Pro, Oracle VirtualBox, and Jira Service Management

Skills Demonstrated

- Active Directory User Administration
 - Password Resets
 - Account Unlocks
 - User Provisioning
 - NTFS & Share Permissions
 - Group Membership Management
 - Windows Domain Administration
 - Help Desk Ticket Documentation
 - Jira Service Management
 - Customer Communication
 - Windows Troubleshooting
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8. Ticket 5 – Domain Join Troubleshooting
9. Skills Demonstrated

Lab Environment.

This home lab simulates a small business Active Directory environment. The lab was created to practice common Tier 1 and Tier 2 IT support tasks including password resets, account management, file permissions, user provisioning, and workstation domain administration. All incidents were tracked through Jira Service Management to simulate a production help desk workflow.

Environment

Component	Description
Hypervisor	Oracle VirtualBox
Server OS	Windows Server 2022
Client OS	Windows 10 Pro
Domain	lab.local
Domain Controller	DC01
Client	CLIENT01
Ticketing	Jira Service Management

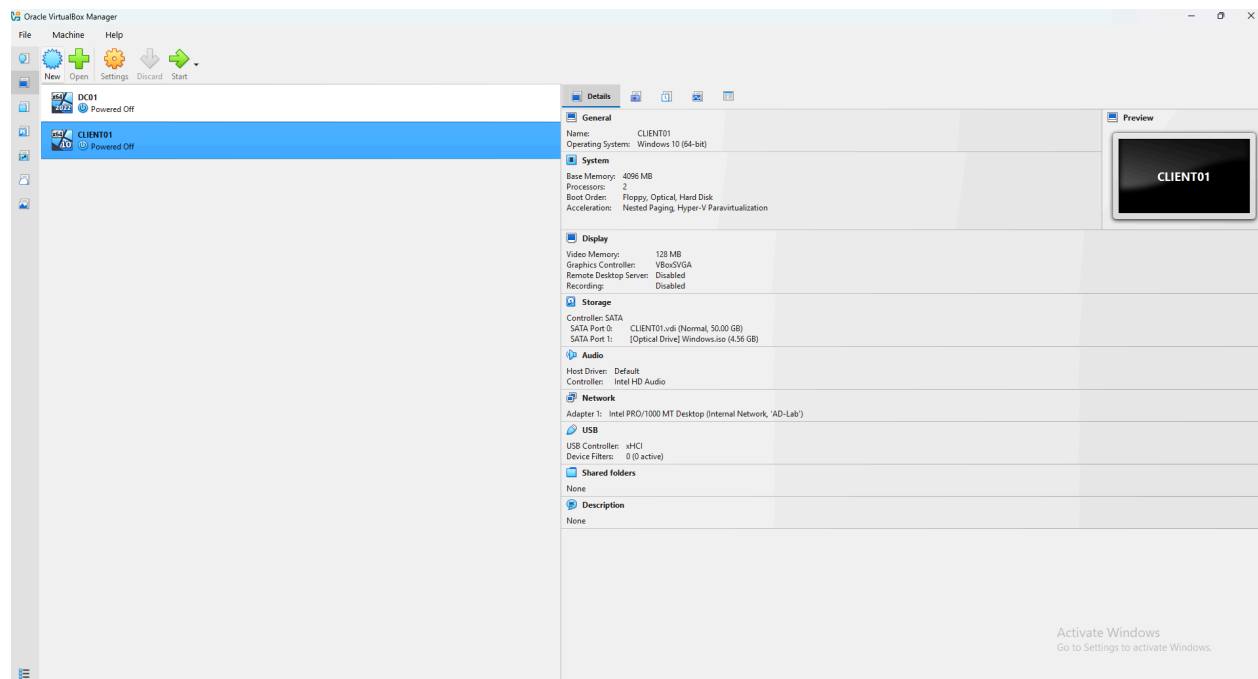


Figure 1.1. VirtualBox Setup with both Domain Controller and Client VMs

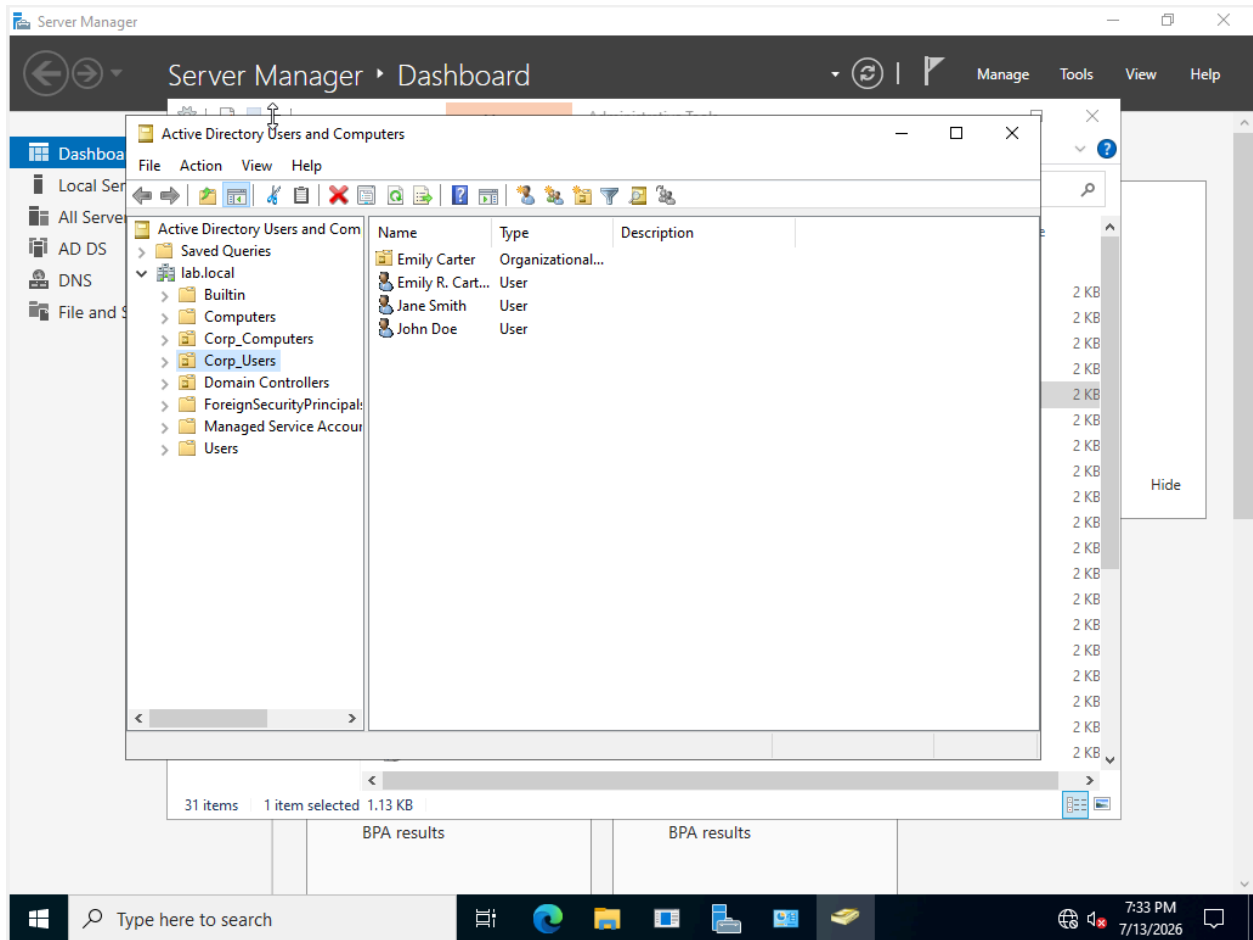


Figure 1.2. Server Manager / Active Directory

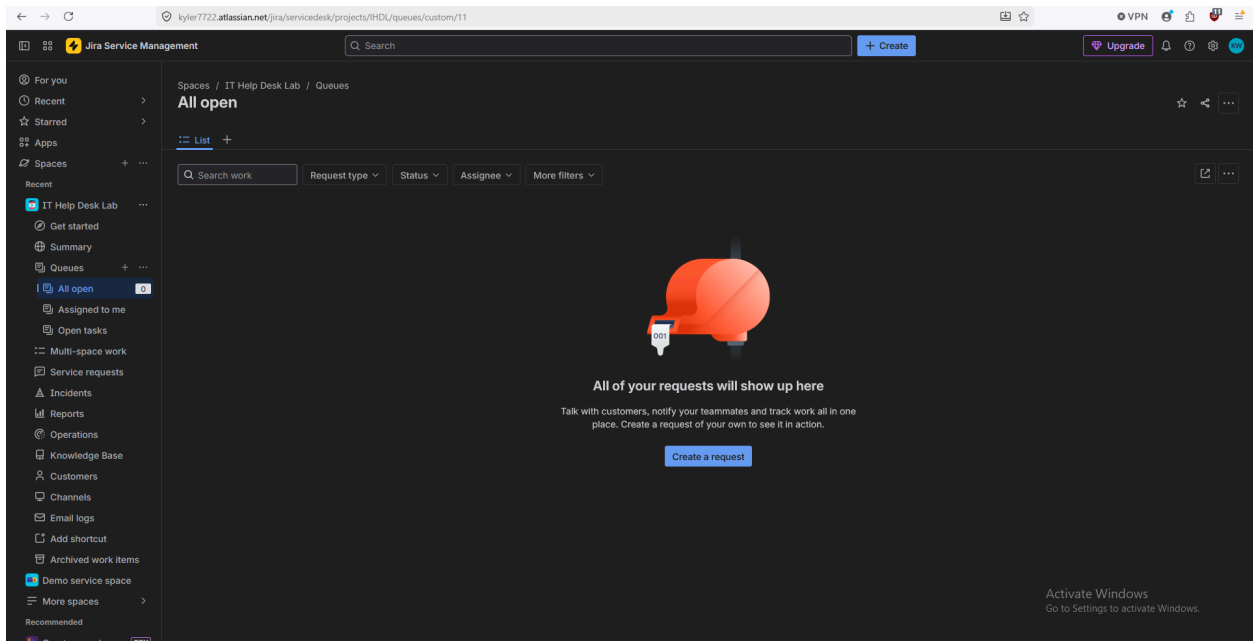


Figure 1.3. Jira Dashboard

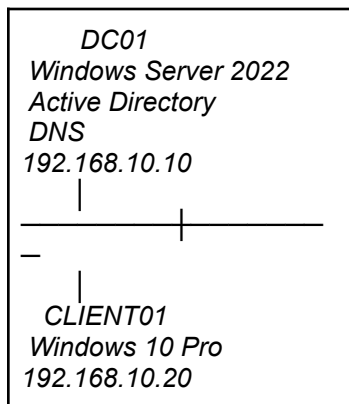


Figure 1.4. Network Diagram for servers

Ticket 1 – Password Reset

Overview

Issue Type: Password Reset

Priority: Medium

System(s): Active Directory, Windows Server 2022, Jira Service Management

Status: Resolved

Scenario

A user contacted the IT Help Desk after forgetting their domain password and was unable to access their workstation. After verifying the user's identity (simulated), the account was located in Active Directory and the password was reset. The user was required to change the password at the next logon.

[Help Center](#) / [IT Help Desk Lab](#)

IT Help Desk Lab

Welcome! You can raise a request for IT Help Desk Lab using the options provided.

Contact us about

Common Requests

What can we help you with?

[Edit request type](#)



Get IT help

Get assistance for general IT problems and questions.

Required fields are marked with an asterisk *

Raise this request on behalf of *

Kyler Watson (kyler7722@gmail.com)

Summary *

Forgot Password

Description

Normal text **B** *I* ... A

Hi, my name is John Doe (jdoe). I have forgotten my password and would like a reset. Thanks!

Figure 2.1. ticket created by client

Troubleshooting Process

1. Reviewed the Jira ticket submitted by the user.
2. Verified the user's identity (simulated).
3. Opened Active Directory Users and Computers.
4. Located the John Doe user account.
5. Reset the account password.
6. Enabled user must change their password at the next logon.
7. Confirmed the user successfully changed the password.
8. Updated the Jira ticket with internal notes.
9. Closed the ticket.

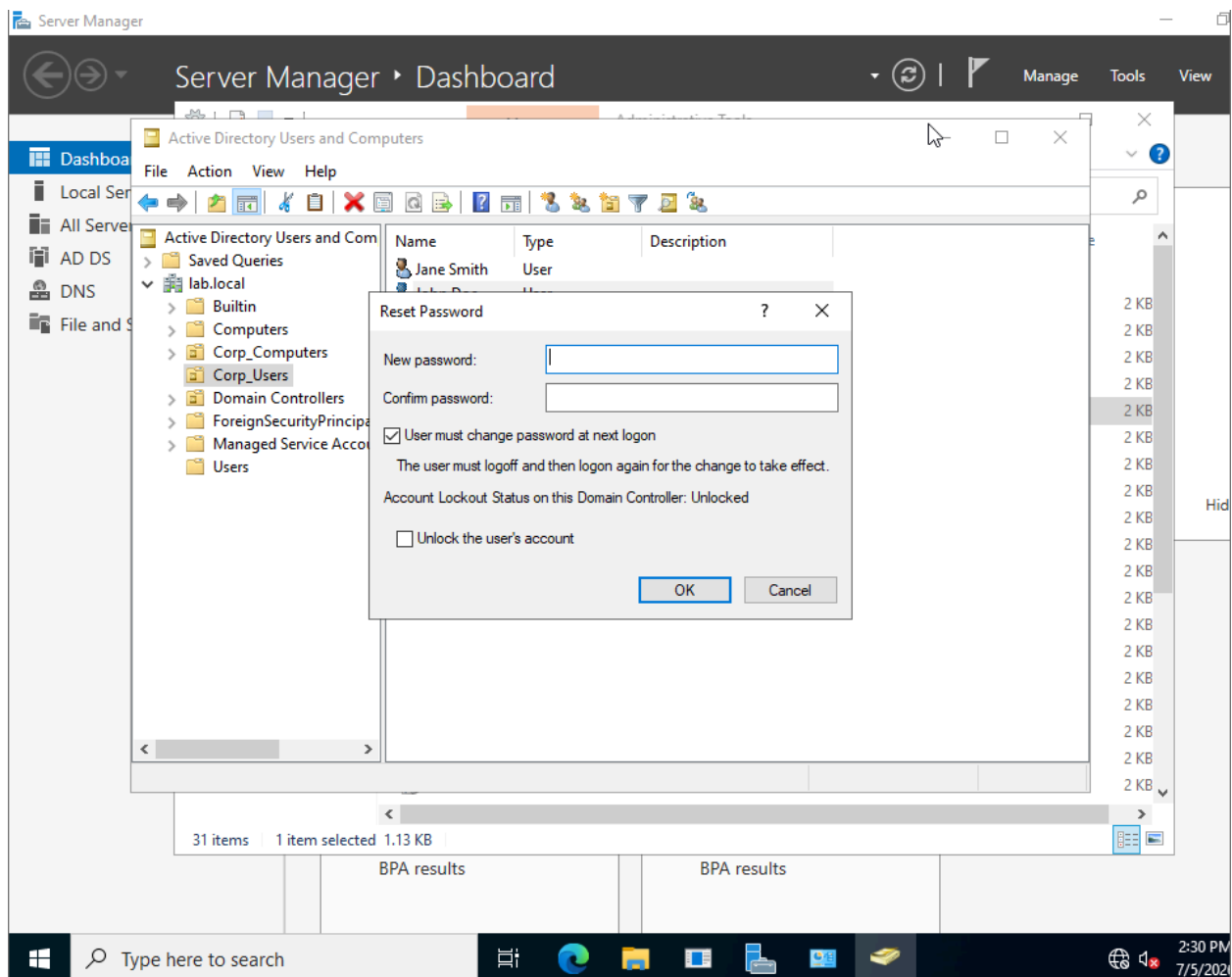


Figure 2.2. found user and gave a temporary password to reset

Resolution

The password was successfully reset in Active Directory. The user changed the temporary password during the next login and regained access to the account. No additional issues were identified.

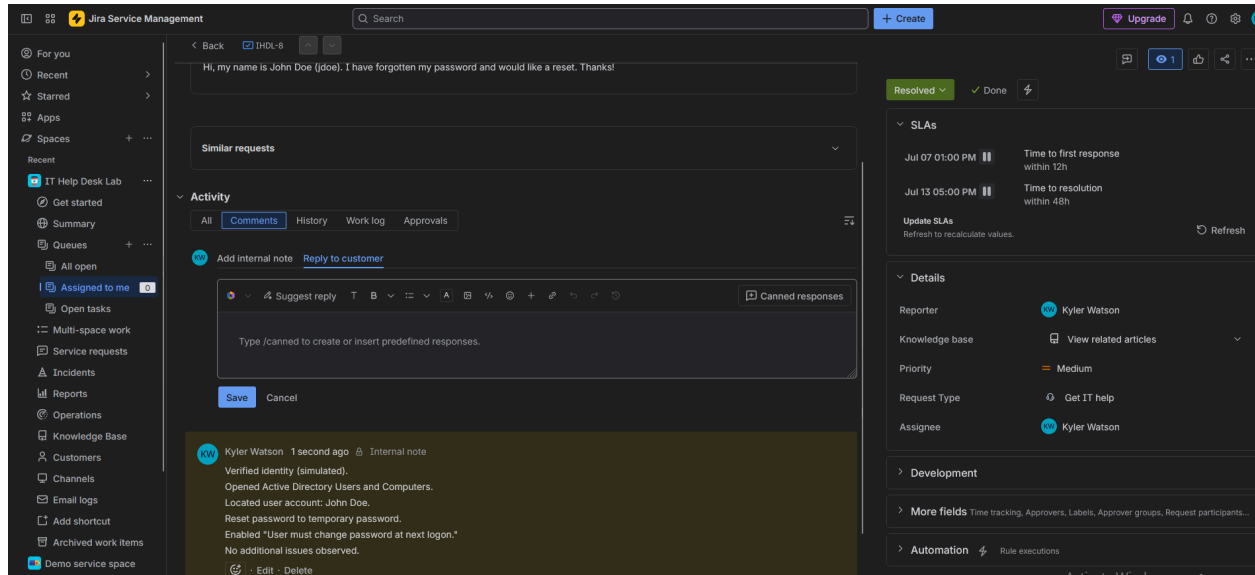


Figure 2.3. Submitted ticket as done

Skills Demonstrated

- Active Directory User Management
- Password Administration
- Windows Server 2022
- User Authentication
- Identity Verification
- Jira Service Management
- Help Desk Documentation

Ticket 2 – Account Lockout Resolution

Overview

Issue Type: Account Lockout

Priority: Medium

System(s): Active Directory, Windows Server 2022, Jira Service Management

Status: Resolved

Scenario

A user contacted the IT Help Desk after being locked out of their Active Directory account due to multiple unsuccessful login attempts. After verifying the user's identity (simulated), the account lockout was confirmed in Active Directory. The account was unlocked, the user was provided with a temporary password, and successful authentication was verified before closing the ticket.

The screenshot displays a Jira Service Management ticket interface. At the top, a blue header bar contains the text "Help Center" on the left and a search icon and a user profile icon (labeled "KW") on the right. Below the header is a blue banner image. The main content area has a breadcrumb trail: "Help Center / IT Help Desk Lab / IHDL-9". The ticket title is "Account locked after failed login attempts".

The ticket details show it was raised by "Kyler Watson" on "Today 3:31 PM". The description reads: "Hi, my name is John Doe (jdoe). My account got locked out due to too many password attempts, can you unlock it and reset my password? Thanks!". Below the description is an "Activity" section with a comment input field labeled "Add a comment".

On the right side, there is a "View work item" link. The "Status" is "WAITING FOR SUPPORT". Below this are several action buttons: "Notifications on", "Escalate", "Cancel request", and "Resolve this issue". The "Request type" is "Get IT help". The "Shared with" section lists "Kyler Watson" as the creator and includes a "Share" button.

At the bottom of the interface, there is a footer that says "Powered by Jira Service Management" and a Windows watermark that says "Activate Windows Go to Settings to activate Windows."

Figure 3.1. Ticket created from client

Troubleshooting Process

1. Reviewed the account lockout request submitted through Jira Service Management.
2. Verified the user's identity (simulated).
3. Opened Active Directory Users and Computers.
4. Located the affected user account.
5. Confirmed the account was locked due to multiple failed login attempts.
6. Unlocked the user account.
7. Reset the user's password to a temporary password.
8. Instructed the user to sign in and change the password.
9. Verified the user successfully authenticated to the domain.
10. Updated the Jira ticket with the investigation and resolution details.
11. Closed the ticket.

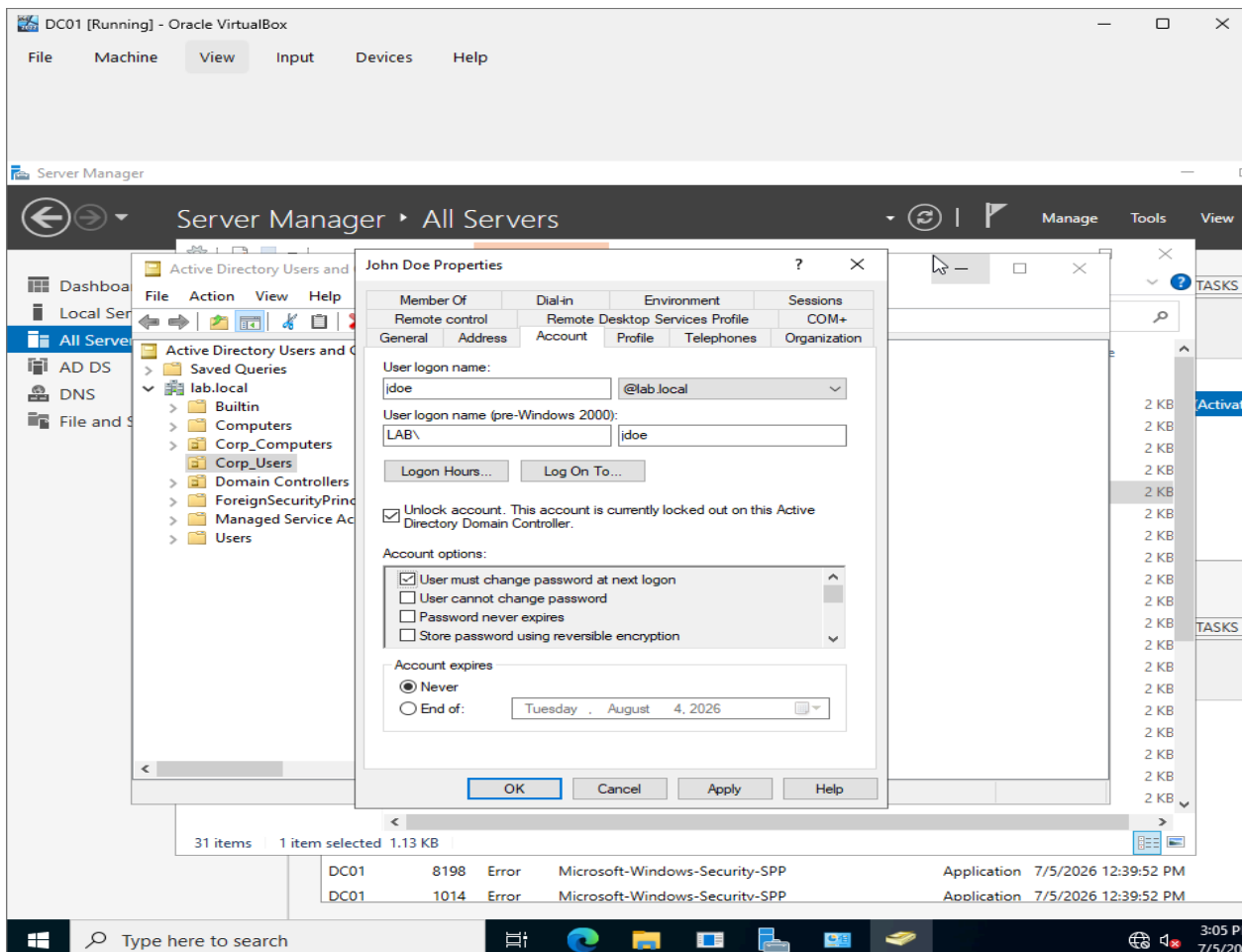


Figure 3.2. found users account, unlocked it and gave a temporary password in order to rest it

Resolution

The user account was successfully unlocked in Active Directory, a temporary password was assigned, and the user regained access to the domain after changing the password during login.

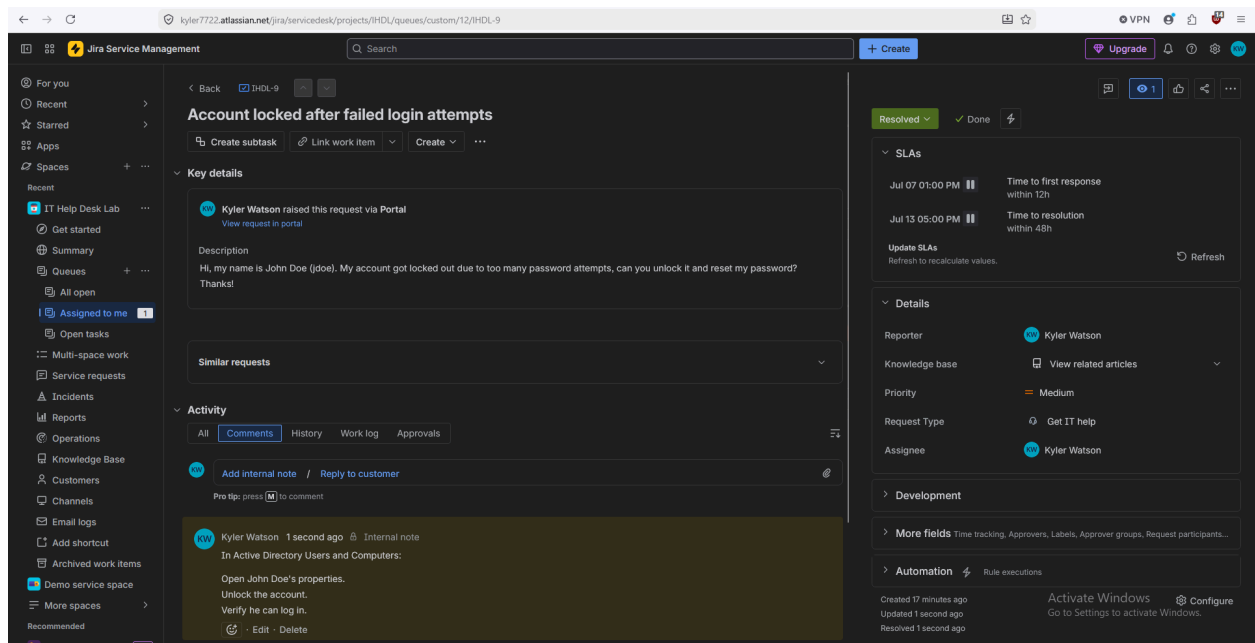


Figure 3.3. Marked ticket as done

Ticket 3 – Shared Folder Access Troubleshooting

Overview

Issue Type: Shared Folder Access / Permissions

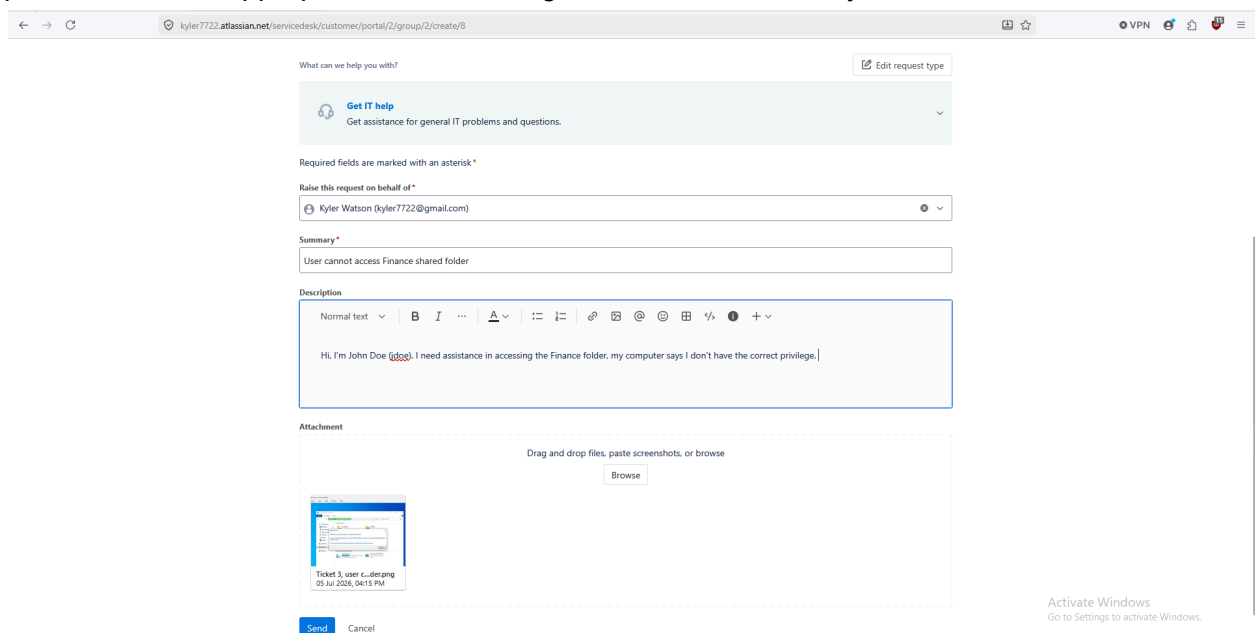
Priority: Medium

System(s): Active Directory, Windows Server 2022, SMB File Sharing, NTFS Permissions, Jira Service Management

Status: Resolved

Scenario

A user contacted the IT Help Desk after receiving an "Access is Denied" message when attempting to access the Finance shared folder on the organization's file server. An investigation was performed to determine whether the issue was related to network connectivity, shared folder permissions, or Active Directory security group membership. After reviewing the user's permissions, the appropriate access was granted and functionality was verified.



The screenshot shows a web browser window with the URL `kyler7722.atlassian.net/servicedesk/customer/portal/2/group/2/create/8`. The page is titled "What can we help you with?" and includes a "Get IT help" button. Below this, there is a section for "Required fields are marked with an asterisk *". The "Raise this request on behalf of *" field is set to "Kyler Watson (kyler7722@gmail.com)". The "Summary *" field contains the text "User cannot access Finance shared folder". The "Description" field has a rich text editor with the text "Hi, I'm John Doe (jdoe). I need assistance in accessing the Finance folder, my computer says I don't have the correct privilege." The "Attachment" section shows a file named "Ticket 3_user_c_creating.png" with a timestamp of "09 Jul 2020, 04:13 PM". The form has "Send" and "Cancel" buttons at the bottom. A "VPN" icon is visible in the top right corner of the browser window.

Figure 4.1. Ticket created by client

Troubleshooting Process

1. Reviewed the incident submitted through Jira Service Management.
2. Confirmed the user received an Access Denied message when attempting to access the Finance shared folder.
3. Verified that the Finance shared folder existed on the Domain Controller.
4. Reviewed both NTFS and Share permissions assigned to the folder.
5. Verified the user's Active Directory group membership.
6. Added the user to the appropriate security group authorized to access the Finance share.
7. Had the user sign out and back into the workstation to refresh security group membership.
8. Verified successful access to the Finance shared folder.
9. Updated the Jira ticket with investigation notes and resolution details.
10. Closed the incident.

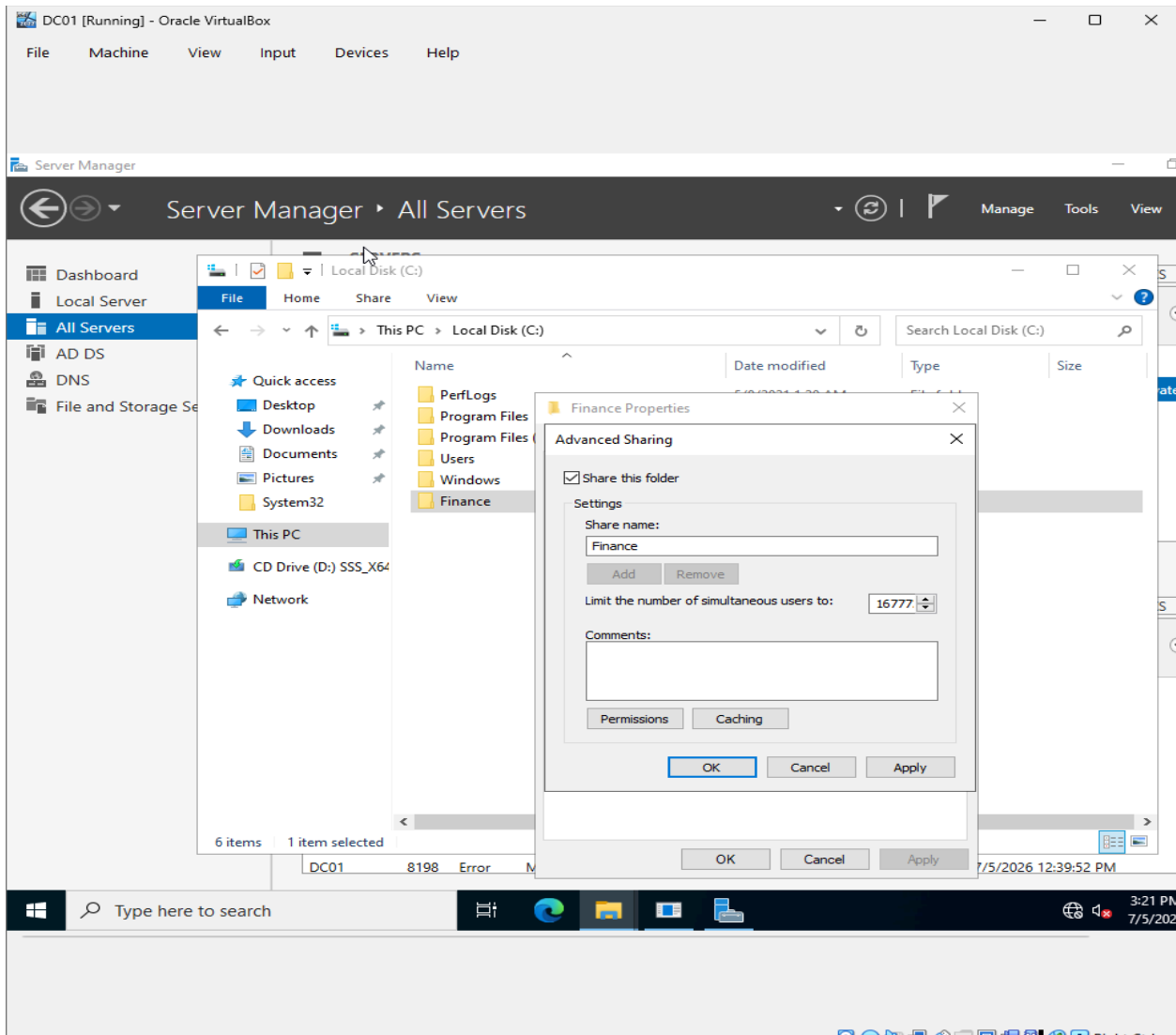
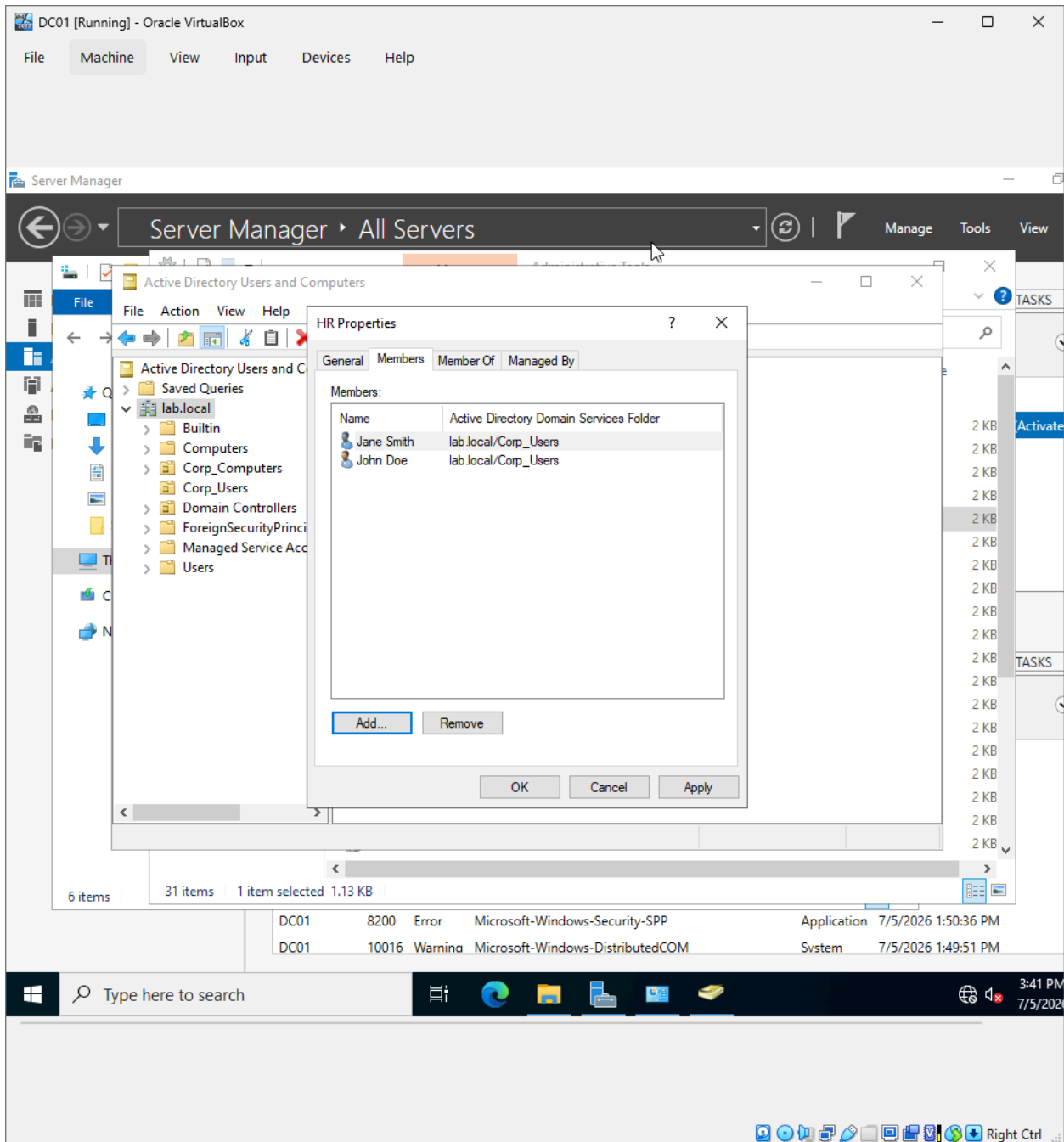


Figure 4.2. Verified that the Finance shared folder existed on the Domain Controller.



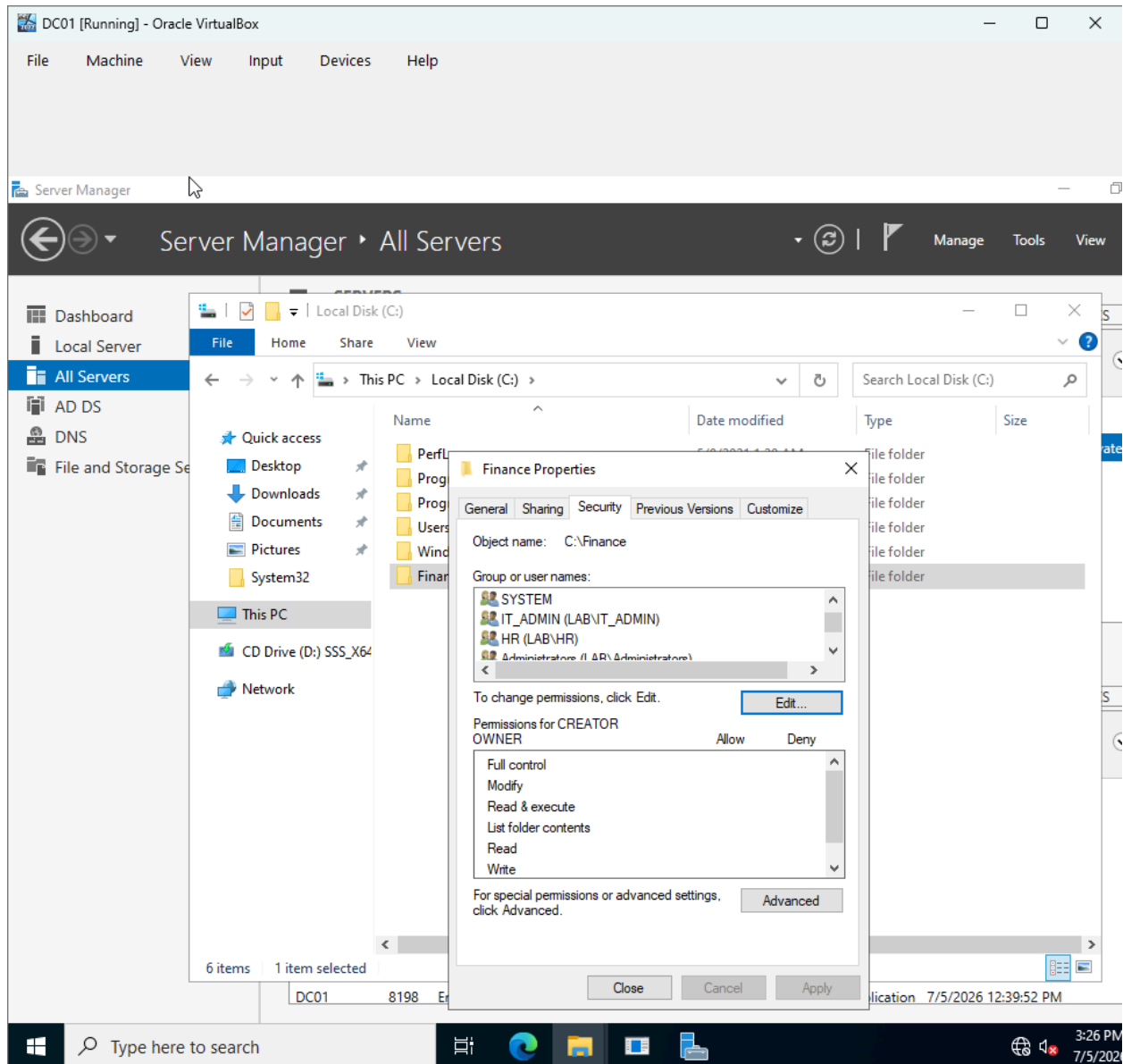


Figure 4.4. Added the user to the appropriate security group authorized to access the Finance share.

Resolution

The user's Active Directory group membership was updated to include the required security group for the Finance shared folder. After the user signed out and back in, Windows refreshed the user's security token and access to the shared folder was successfully restored.

Browser tabs: [IHDL-10] User cannot access Finance, User cannot access Finance, Home Lab Documentation

URL: kyler7722.atlassian.net/jira/servicedesk/projects/IHDL/queues/custom/1

Buttons: + Create, Upgrade, Search, + Create, Upgrade, ?

Kyler Watson raised this request via Portal
[View request in portal](#)

Description
Hi, I'm John Doe (jdoe). I need assistance in accessing the Finance folder, my computer says I don't have the correct privilege.

Attachments 1

Ticket 3, user c..., der.png
05 Jul 2026, 04:15 PM

Similar requests

Activity
All Comments History Work log Approvals

Add internal note / **Reply to customer**
Pro tip: press **M** to comment

Kyler Watson 1 second ago Internal note
User successfully accessed the Finance shared folder after group membership was updated and a new logon session was established.
Edit Delete

Kyler Watson now
Your access has been updated. Please sign out of Windows and sign

Time to close after resolution
within 60h
Today 04:31 PM ✓

Time to first response
within 12h
Today 04:31 PM ✓

Time to resolution
within 48h

Update SLAs
Refresh to recalculate values. Refresh

Details

Reporter
KW Kyler Watson

Knowledge base
View related articles

Priority
Medium

Request Type
Get IT help

Assignee
KW Kyler Watson

Development

Automation Rule executions

Figure 5.5. Marked ticket as done

Ticket 4 – New Employee Account Provisioning

Overview

Issue Type: User Account Provisioning / New Employee Onboarding

Priority: Medium

System(s): Active Directory, Windows Server 2022, Active Directory Users and Computers, Security Groups, SMB File Sharing, Jira Service Management

Status: Resolved

Scenario

The Human Resources department submitted a service request to create a domain account for a newly hired employee. The employee required a standard Active Directory account, assignment to the appropriate security group, and access to departmental resources before their first day of work. The account was created, appropriate permissions were assigned, and successful authentication and resource access were verified.

Troubleshooting / Implementation Process

1. Reviewed the onboarding request submitted through Jira Service Management.
2. Opened Active Directory Users and Computers.
3. Created a new domain user account for Emily Carter.
4. Assigned the account to the appropriate Organizational Unit.
5. Added the user to the HR security group.
6. Configured a temporary password.
7. Enabled User must change their password at the next logon.
8. Logged into CLIENT01 using the newly created account.
9. Verified successful authentication.
10. Confirmed the user could access the Finance shared folder.
11. Updated the Jira ticket with implementation details.
12. Closed the request after successful verification.

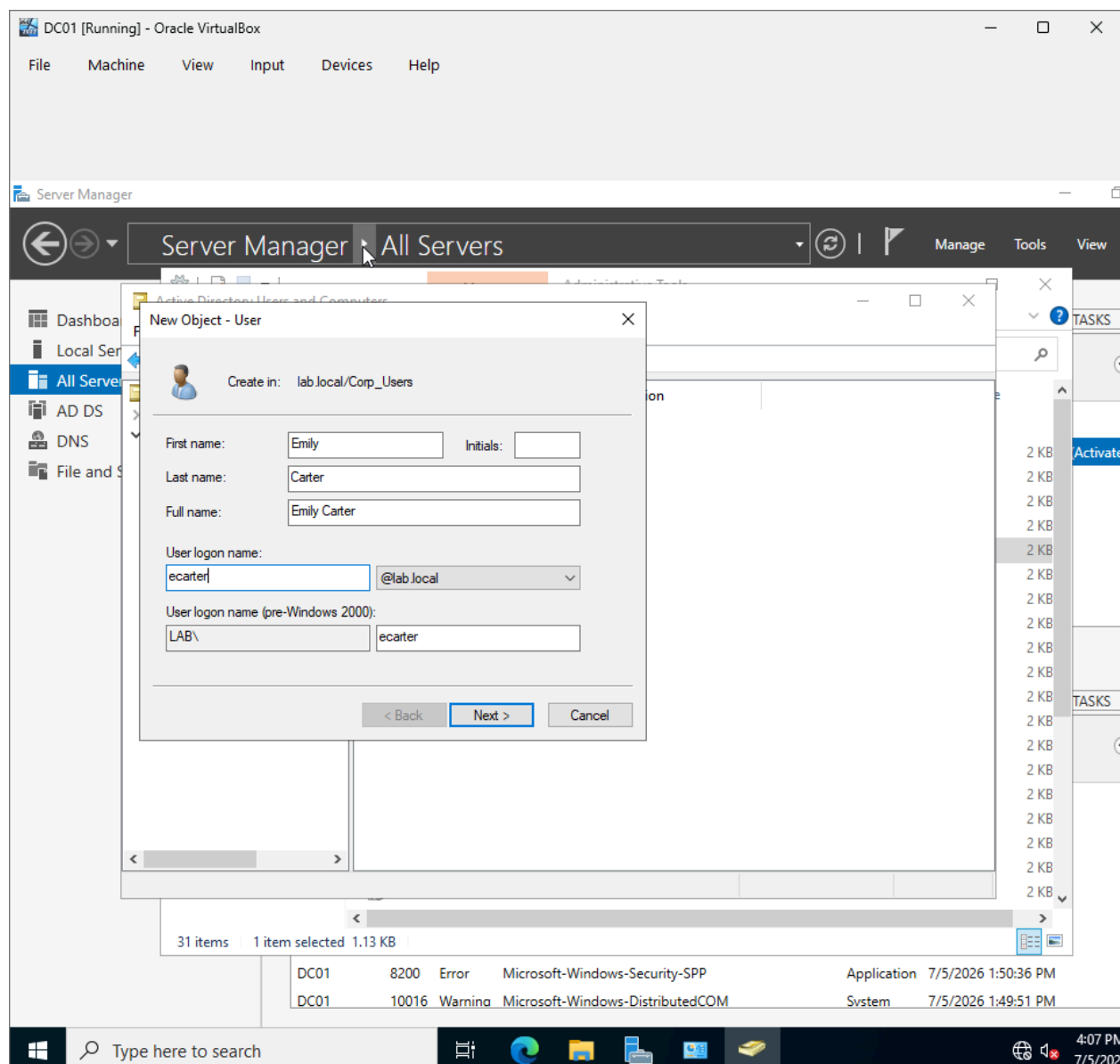


Figure 5.2. Created new account in Active Directory

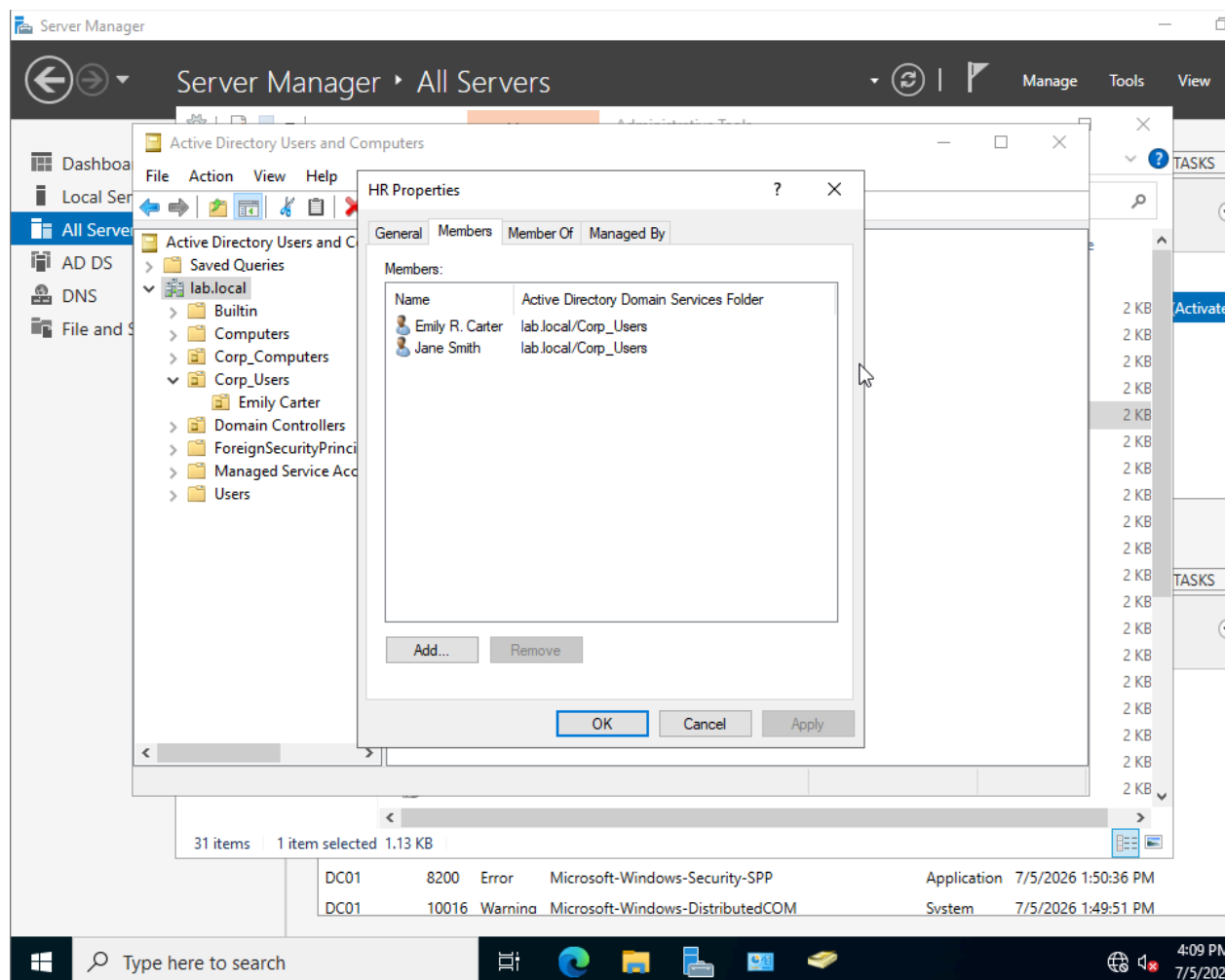


Figure 5.3, added new user to HR group

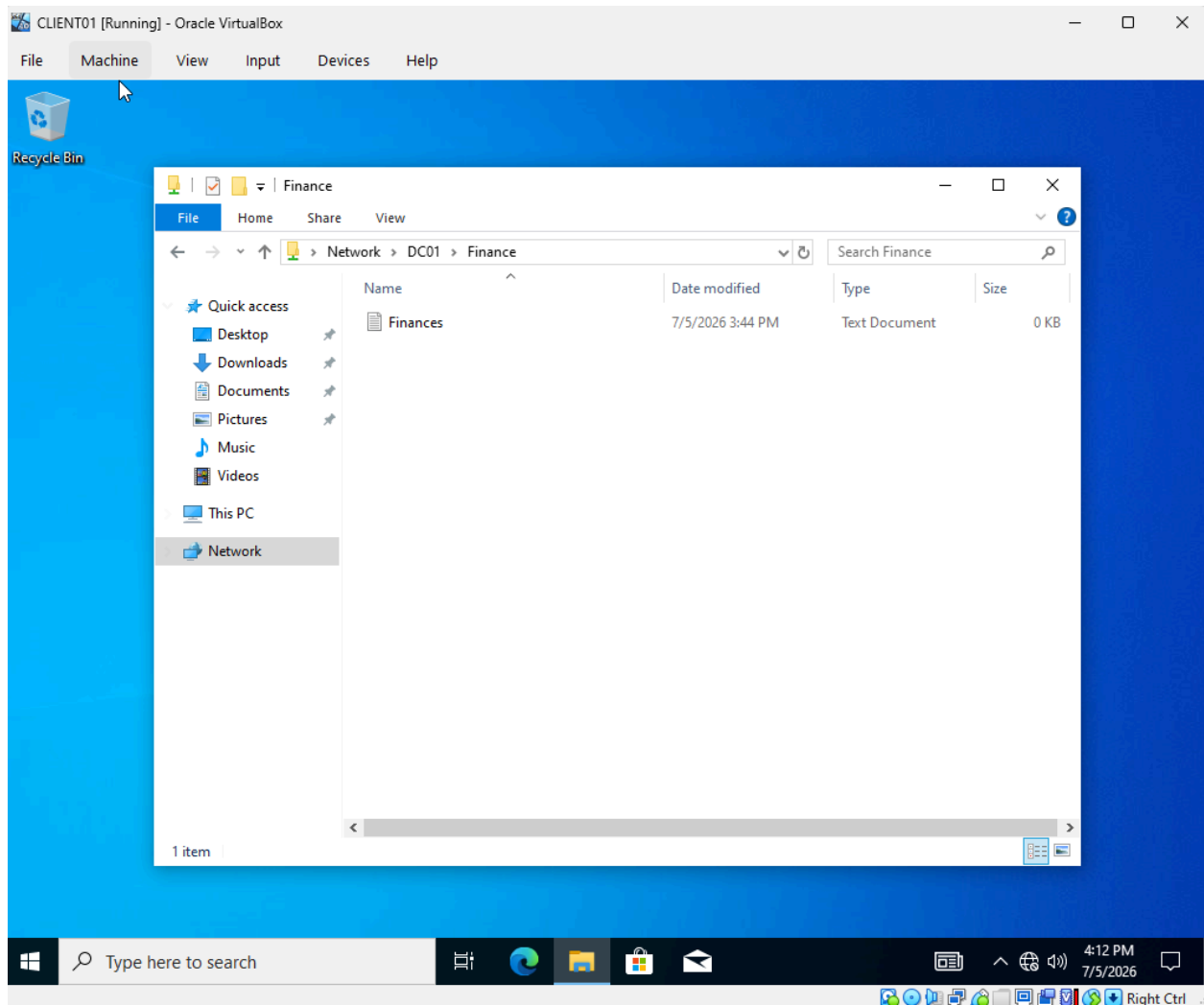


Figure 5.4. New user has access to Finance Folder

Resolution

A new Active Directory user account was successfully created and configured according to organizational standards. The user was assigned to the appropriate security group, required to change the temporary password during first logon, and granted access to the Finance shared folder. Authentication and resource access were verified before closing the request.

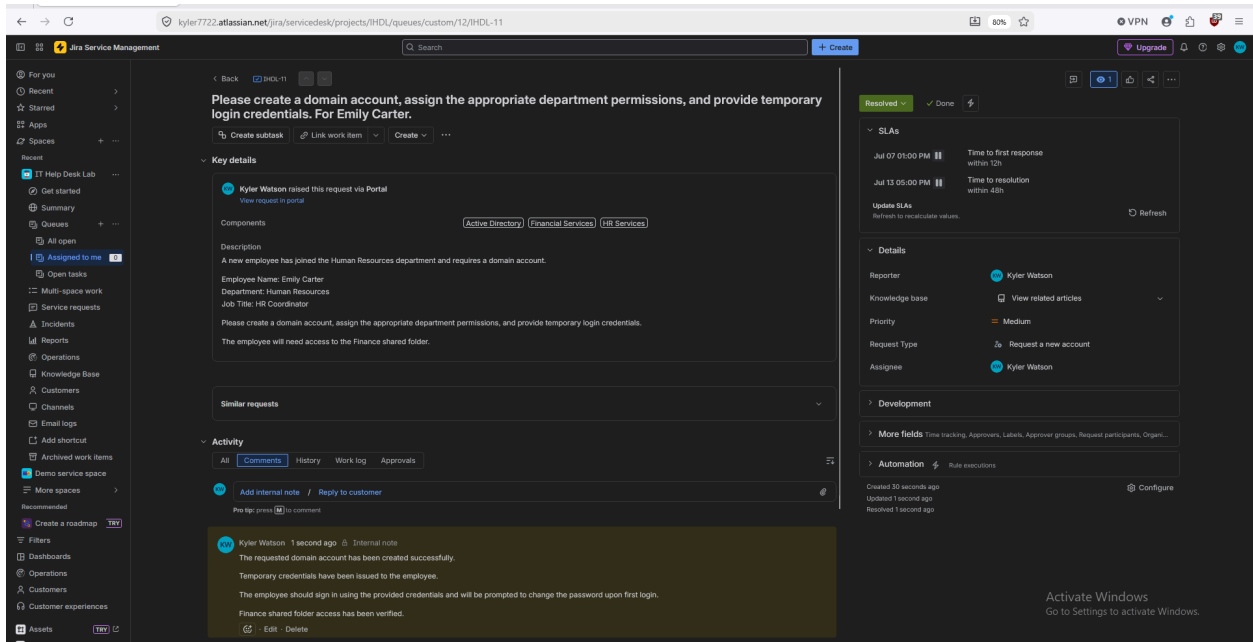


Figure 5.5. Marked ticket as done

Lessons Learned

This project provided practical experience administering a Windows Active Directory environment while simulating common Help Desk workflows.

Key lessons included:

- Active Directory relies heavily on proper DNS configuration.
- Security groups simplify permission management by assigning access based on organizational roles rather than individual users.
- Both NTFS and Share permissions must be configured correctly to secure shared resources.
- Structured troubleshooting helps isolate problems efficiently and reduces resolution time.
- Clear documentation within a ticketing system is just as important as the technical solution itself.

Completing simulated support requests from initial user submission through final verification reinforced the importance of communication, documentation, and validation throughout the incident management process.

Skills Demonstrated

Windows Server Administration

- Windows Server 2022
- Server Manager
- Active Directory Domain Services
- DNS

Active Directory

- User Management
- Organizational Units
- Security Groups
- Password Administration
- Account Lockout Recovery
- User Provisioning

File Services

- SMB File Sharing
- NTFS Permissions
- Share Permissions

Networking

- DNS
- Domain Join
- Windows Authentication

Help Desk

- Incident Documentation
 - Customer Communication
 - Root Cause Analysis
 - Jira Service Management
-

Conclusion

This project successfully simulated a small enterprise Windows environment using Windows Server 2022, Active Directory Domain Services, DNS, VirtualBox, and Jira Service Management.

Through the completion of multiple Help Desk scenarios, I gained practical experience managing user accounts, troubleshooting authentication issues, configuring file permissions, provisioning new employees, and documenting support incidents using a structured workflow.

This lab strengthened my understanding of enterprise Windows administration while reinforcing the importance of technical documentation, communication, and systematic troubleshooting.